

SECURITY MONITORING SYSTEM ON AWS

Prepared in the partial fulfillment of the Summer
Internship Program on AWS



Under the guidance of
Mrs. Sumana Bethala, APSSDC
Mr. Anil Kumar, APSSDC

Group Members

- 1.23MH1A0430 – Kalagara Kusuma
- 2.23MH1A4902 – Aketi Lakshmi Srujana
- 3.23MH1A4911 – Chevvakula Sireesha
- 4.23MH1A4951 – Addepalli Ramya Haripriya
Varshini
- 5.23MH1A4953 – Rayudu Satya Sri

Abstract

- The Security Monitoring System is a smart solution that uses sensors, cameras, and cloud technology to detect and alert users about security threats in real-time. It ensures safety through automated monitoring, reducing manual effort and enabling quick response to incidents.

INTRODUCTION

- Ever wondered how we can keep tabs on who's accessing our most sensitive data in AWS? Whether it's API keys, database credentials, or other critical secrets, every access poses a potential security risk. That's why organizations invest heavily in robust monitoring systems to track and alert on any unusual activity
- In this group project, we'll build our own powerful monitoring system using AWS. Protecting data is an essential skill for roles like Security Engineer, DevOps Engineer, and Systems or Cloud Administrator, and by working together we'll gain hands-on experience in designing, deploying, and maintaining an end-to-end audit and alerting solution.

Problem Statement

- Security breaches are increasing in cloud infrastructure.
- Traditional monitoring systems are insufficient for dynamic, scalable AWS environments.
- Need for real-time, automated, and scalable security monitoring.

Solution Overview

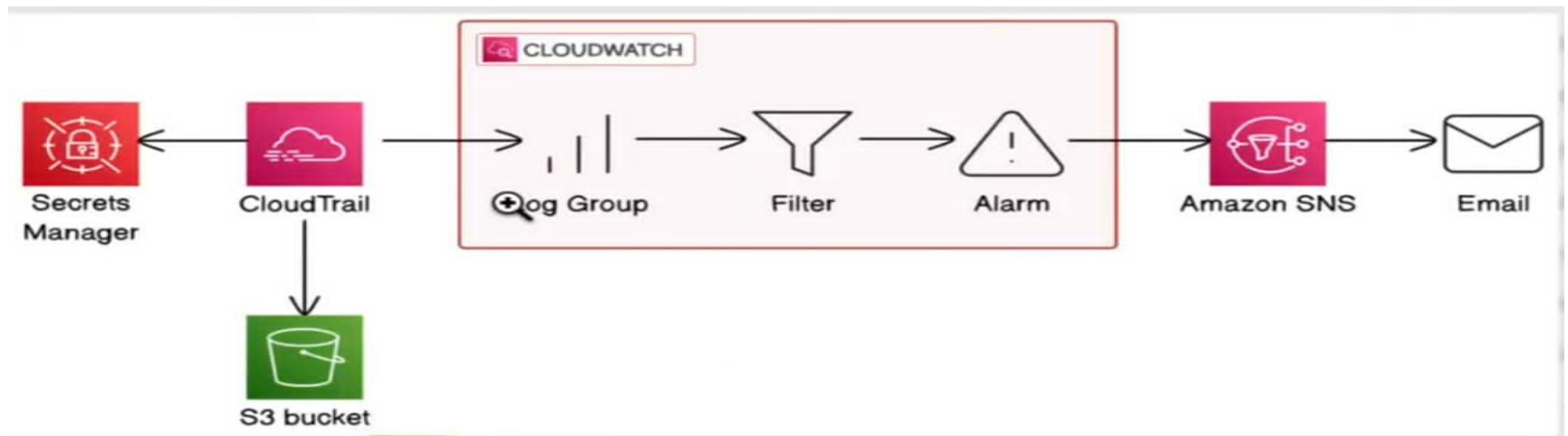
- Built a Security Monitoring System on AWS.
- Uses AWS native services to monitor, alert, and log security threats.
- Designed for scalability, automation, and minimal manual intervention.

Key Components

- Amazon CloudWatch: Logs and monitors metrics/events.
- AWS CloudTrail: Tracks API activity.
- Amazon GuardDuty: Detects threats using ML and threat intelligence.
- AWS Config: Monitors compliance and resource configuration changes.
- SNS & Lambda: Triggers alerts and custom workflows.

- * Set up AWS CloudTrail to track secret access events.
- * Use AWS CloudWatch to log access attempts and trigger notifications.
- * Create SNS alerts to get notified when your secrets are accessed.

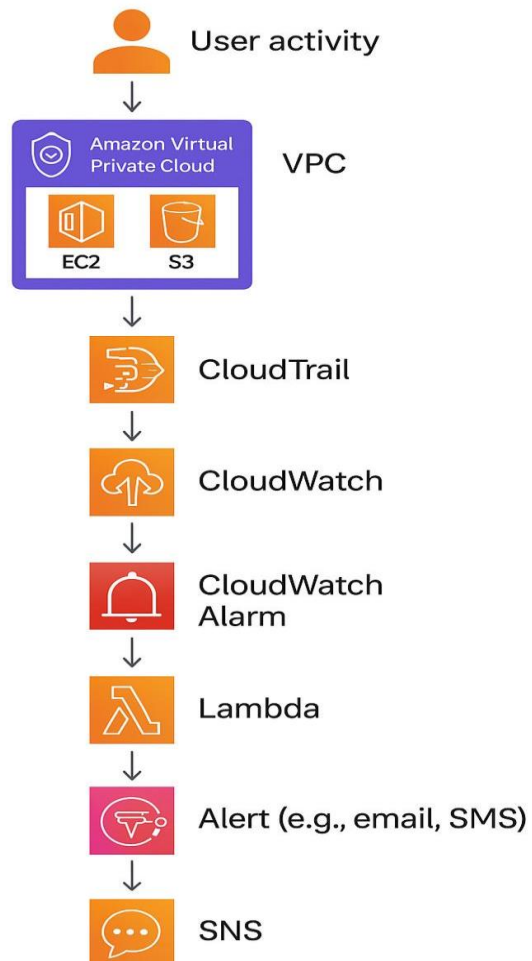
Build a second notification system and compare which approach delivers better security alerts.



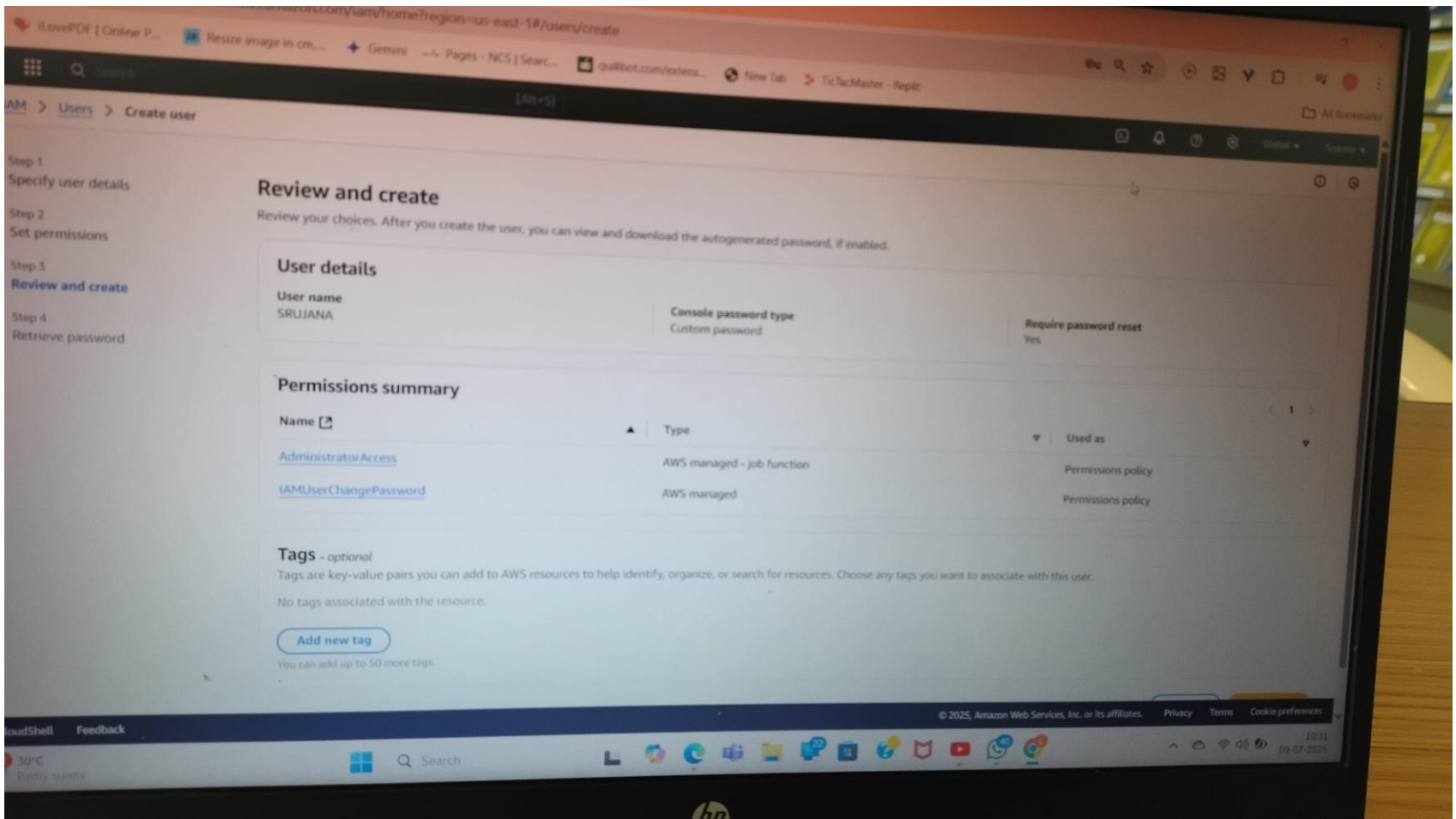
Key Features

- Real-time anomaly detection
- Auto-notifications via SNS
- Compliance monitoring via AWS Config
- Centralized log collection
- Scalable & serverless setup

Flowchart Diagram



Results/Demo



Results/Demo

The screenshot displays the AWS Secrets Manager console for the 'TopSecretInfo' secret in the 'us-east-1' region. The secret details show an encryption key of 'aws/secretsmanager' and a description: 'Secret created for security monitoring on aws'. Below the console, a CloudShell terminal window is open, showing the command to retrieve the secret value and its output.

Secret details

- Encryption key: [aws/secretsmanager](#)
- Secret name: [TopSecretInfo](#)
- Secret description: Secret created for security monitoring on aws

CloudShell

```
us-east-1 +  
~ $ aws secretsmanager get-secret-value --secret-id "TopSecretInfo" --region us-east-1  
{  
  "ARN": "arn:aws:secretsmanager:us-east-1:238576302453:secret:TopSecretInfo-fr06e0",  
  "Name": "TopSecretInfo",  
  "VersionId": "fbac86e7-d4dc-4b25-b478-14fcb8d8824a",  
  "SecretString": "{\\\"The API Key Is\\\":\\\"Build a security monitoring on aws123\\\"}",  
  "VersionStages": [  
    "AWSCURRENT"  
  ],  
  "CreateDate": "2025-07-09T09:05:11.649000+00:00"  
}  
~ $
```

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Results/Demo



Simple Notification Service

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`arn:aws:sns:us-east-1:238576302453:SecurityAlarms:d13a3ac3-3cbc-40de-86b7-7410ff4b95ef`

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Learnings and Outcomes

- Gained hands-on experience with AWS monitoring tools
- Understood the importance of continuous compliance
- Learned to build scalable, event-driven architecture
- Ready to apply this in real-world enterprise security projects

Conclusion

- This project showcases how AWS can be used to build robust, automated security monitoring without external tools. Future enhancements could include integrating Lambda for automated remediation, visual dashboards using QuickSight, and deeper log analytics with OpenSearch or Athena.

Thank You